

CSE 342
Statistical Machine Learning
Assignment - 1

Anushk Kumar
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Code Explanation

The code is divided into functional components, with each function performing a specific task in the overall pipeline. Experimentation is performed by setting a Random Seed (**2023115**) to ensure reproducible results across iterations.

- *load_dataset()*: This function loads the MNIST dataset by taking file paths of labels and images stored inubyte format and performs processing using struct, and returns numpy arrays of images and labels.
- *preprocess()*: This function takes input of numpy arrays of images and flattens the 2D image array into a 1D array and normalises the values into the range [0,1], and returns the normalised image array
- *sample_data()*: This function is used to sample data points with count specified as sample_size and after sampling adds to individual image array and label array and shuffles them using numpy permutation and returns the shuffled numpy arrays of the data points.
- *filter012()*: This function filters out training images and labels that belong to classes 0, 1, and 2. And returns the updated subset of images and labels.
- *MLEstimate()*: This function estimates the parameters for a Gaussian-based classifier, like Mean, Covariance, and prior, from the input training images and labels, and returns a list of ordered parameters (mean, cov, prior) for each class c.
- *LDA()*: This implements Linear Discriminant Analysis for classification. It assumes all classes share a single "pooled" covariance matrix and calculates a discriminant score for each class based on the distance between the test image and the class mean; the class with the highest score is chosen as the prediction.

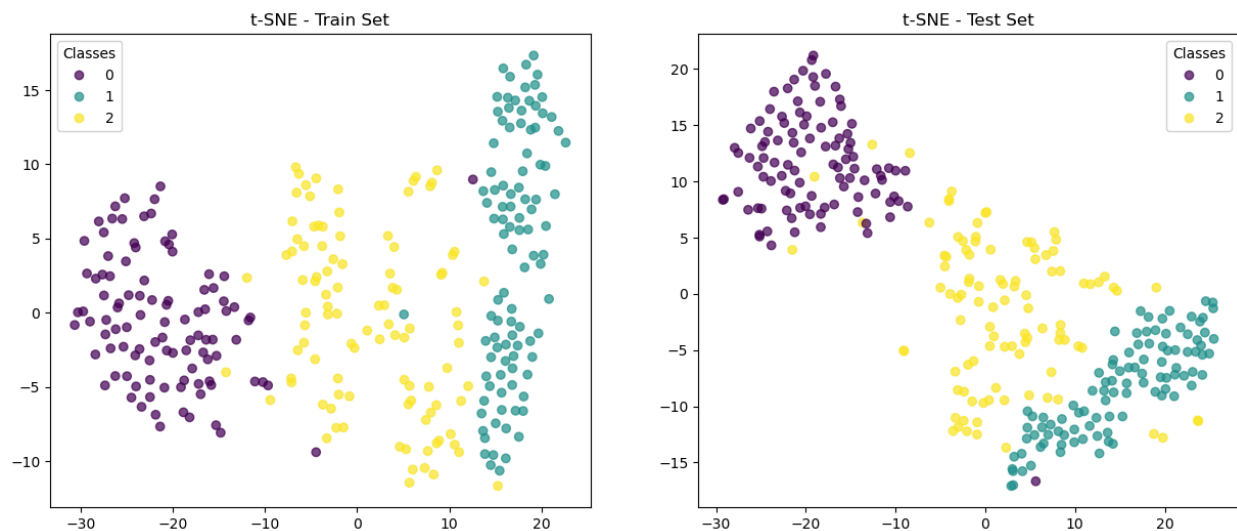
- **QDA():** This implements Quadratic Discriminant Analysis. Unlike LDA, it uses a unique covariance matrix for each class, allowing more flexible decision boundaries; it computes the log-determinant and inverse of each class's covariance to determine the most likely class for a given image.

The results of the experimentation were as follows:

```
LDA Discriminant values for first test sample: ['Class 0: -200165.62', 'Class 1: -198906.91', 'Class 2: -188935.35']
LDA Accuracy: 0.8167
QDA Discriminant values for first test sample: ['Class 0: -1171575.61', 'Class 1: -2382029.30', 'Class 2: -774794.10']
QDA Accuracy: 0.9833
```

Visualisation Explanation (t-SNE plots)

The following image shows t-SNE plots of the sampled training and test sets from the MNIST dataset. This was generated using the sklearn TSNE algorithm. *visualise()* function creates a 2D map of the high-dimensional image data using the t-SNE algorithm. It generates two side-by-side scatter plots (one for training and one for testing) where the colors represent the digit classes, allowing you to see how well the classes naturally separate in space.



Here in these plots, we can effectively infer

In your generated output, the **Train Set** and **Test Set** plots show three distinct clusters:

- **Class 0 (Digit 0):** Usually forms a very tight, separate cluster because its shape is circular and distinct from 1s and 2s.
- **Class 1 (Digit 1):** Typically forms a linear or elongated cluster.
- **Class 2 (Digit 2):** Forms the third cluster.